Questions by topic: Sampling techniques & measuring abiotic factors - Set 1 (85 marks)

Answer all questions in the spaces provided.

0 1 A student used 0.5 m x 0.5 m grid quadrats to survey ground flora in two meadows, one grazed and one ungrazed. Table 1 shows the results for the grazed meadow (10 quadrats).

Table 1

Species	Number of individuals
Ribwort plantain	48
Perennial ryegrass	120
White clover	36
Daisy	24
Creeping buttercup	12

0 1 . 1 Explain why a 0.5 m x 0.5 m quadrat was chosen rather than a 10 cm x 10 cm or 100 m x 100 m quadrat.

[2 marks]

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0 1 . 2 Calculate Simpson's Index of Biodiversity (D) for the grazed meadow using D = N(N-1) / Σn(n-1). Show your working.

[3 marks]

Answer

0 1 . 3 The ungrazed meadow had D = 1.8 with the same number of species. Explain what this shows.

[2 marks]

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0 1 . 4 Explain why a point quadrat would be better than a frame quadrat in tall vegetation.

[2 marks]

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0 1 . 5 Give **one** limitation of using the DAFOR abundance scale.

[1 mark]

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0 2 A student compared the invertebrates in a stream above and below a sewage outfall using kick sampling.

0 2 . 1 Describe how kick sampling is carried out so that the samples from the two sites are comparable.

[3 marks]

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0 2 . 2 Give **three** limitations of kick sampling.

[3 marks]

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0 2 . 3 Explain how a Surber sampler improves on kick sampling.

[2 marks]

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0 2 . 4 Explain why the invertebrates found give a better indication of long-term sewage pollution than a single measurement of dissolved oxygen.

[2 marks]

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0 3 . 1 Complete Table 1 to show a suitable sampling technique for each group of organisms and one limitation of the technique.

[6 marks]

Table 1

Organisms	Technique	One limitation	
Invertebrates active on the ground surface at night			
Invertebrates living in leaf litter			
Invertebrates on tree branches			

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0 3 . 2 Explain why a light trap cannot be used to prove that a moth species is absent from a site.

[2 marks]

0 3 . 3 Explain how a motor-driven suction sampler can give a population density, whereas a pooter cannot.

[2 marks]

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0 4 A student estimated the population of ground beetles in a field using the Lincoln Index. 60 beetles were caught in pitfall traps, marked with a spot of paint and released. Three days later 45 beetles were caught, of which 9 were marked.

0 4 . 1 Calculate the estimated population size.

[2 marks]

Answer ..... beetles

0 4 . 2 State **four** assumptions of the Lincoln Index method.

[4 marks]

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0 4 . 3 Explain how the estimate would be affected if the paint made the beetles more likely to be eaten by birds.

[2 marks]

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0 4 . 4 Suggest why the second sample was taken after three days rather than the next day or after a month.

[2 marks]

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0 5 . 1 Explain why electronic meters must be calibrated before use and describe how a pH meter is calibrated.

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0 5 . 2 Describe how wind velocity measurements with an anemometer should be standardised.

[3 marks]

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0 5 . 3 Describe how a whirling hygrometer is used to measure humidity.

[2 marks]

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0 5 . 4 Describe **two** methods of measuring nitrate concentration in water.

[2 marks]

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0 6 . 1 Describe how soil texture is determined using a stack of sieves.

[3 marks]

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0 6 . 2 Explain why a soil auger gives a more representative sample than a trowel.

[2 marks]

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0 6 . 3 Explain why the organic matter content will be overestimated if a soil sample is heated above 550 °C.

[2 marks]

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0 6 . 4 Describe how a turbidimeter measures water turbidity and explain how quantitative results are obtained.

[3 marks]

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0 7 A student investigated the effect of pH on the germination of cress seeds.

0 7 . 1 Explain why cress is a suitable species for this investigation but the results may not apply directly to natural ecosystems.

[2 marks]

0 7 . 2 Describe how the student should decide the range of pH values to test.

[2 marks]

0 7 . 3 Give **three** variables that should be standardised and explain how long the experiment should continue.

[4 marks]

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0 7 . 4 The student wants to compare the effect of nutrient concentration on the growth of duckweed. Explain why leaf area is a better measure of growth than fresh mass.

[2 marks]

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0 8 A student is comparing the plant diversity of a grazed and an ungrazed grassland, and estimating the population of grasshoppers in each. In the grazed field 60 grasshoppers were caught, marked and released; three days later 75 were caught, of which 15 were marked.

0 8 . 1 Explain **three** reasons for choosing quadrat positions using random numbers rather than by eye.

[3 marks]

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0 8 . 2 Use the Lincoln index to estimate the grasshopper population of the grazed field.

[2 marks]

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Answer

0 8 . 3 State **three** assumptions that must hold for this estimate to be valid.

[3 marks]

0 8 . 4 Explain how the student should decide the size of quadrat and the number of quadrats to use.

[3 marks]

0 8 . 5 Explain **four** other features of the design that would make the comparison between the two fields more reliable or more valid.

[4 marks]

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END OF QUESTIONS

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